

PUBLICATIONS

14. Tiwari, V; Korol, R; Franco*, I Robust Purely Optical Signatures of Floquet States in Laser-Dressed Crystals. *Phys. Rev. Lett.* **2025**, 135 (18), 186901. DOI: [10.1103/5ywx-7dbs](https://doi.org/10.1103/5ywx-7dbs)
13. Korol*, R; Chen, X; Franco*, I High-Frequency Tails in Spectral Densities. *J. Phys. Chem. A* **2025**, 129 (15), 3587-3596. DOI: [10.1021/acs.jpca.5c00943](https://doi.org/10.1021/acs.jpca.5c00943)
12. Turner*, A C; Korol, R; Bill, M; Stolper, D A Stable isotope equilibria in the dihydrogen-water-methane-ethane-propane system. Part 2: Experimental determination of hydrogen isotopic equilibrium for ethane-H₂ from 30 to 200 °C and propane-H₂ from 75 to 200 °C. *Geochim. Cosmochim. Acta* **2025**, 396, 91-106. DOI: [10.1016/j.gca.2025.02.033](https://doi.org/10.1016/j.gca.2025.02.033)
11. Korol*, R; Turner, A C; Nandi, A; Bowman, J M; Goddard, W A; Stolper, D A Stable isotope equilibria in the dihydrogen-water-methane-ethane-propane system. Part 1: Path-integral calculations with CCSD(T) quality potentials. *Geochim. Cosmochim. Acta* **2025**, 396, 71-90. DOI: [10.1016/j.gca.2025.02.028](https://doi.org/10.1016/j.gca.2025.02.028)
10. Turner, A C; Korol, R; Eldridge, D L; Bill, M; Conrad, M E; Miller, T F; Stolper*, D A Experimental and theoretical determinations of hydrogen isotopic equilibrium in the system CH₄-H₂-H₂O from 3 to 200 °C. *Geochim. Cosmochim. Acta* **2021**, 314, 223-269. DOI: [10.1016/j.gca.2021.04.026](https://doi.org/10.1016/j.gca.2021.04.026)
9. Korol, R; Rosa-Raíces, J L; Bou-Rabee, N; Miller*, T F Dimension-free path-integral molecular dynamics without preconditioning. *J. Chem. Phys.* **2020**, 152 (10), 104102. DOI: [10.1063/1.5134810](https://doi.org/10.1063/1.5134810)
8. Eldridge, D L; Korol, R; Lloyd, M K; Turner, A C; Webb, M A; Miller, T F; Stolper*, D A Comparison of Experimental vs Theoretical Abundances of ¹³CH₃D and ¹²CH₂D₂ for Isotopically Equilibrated Systems from 1 to 500 °C. *ACS Earth Space Chem.* **2019**, 3 (12), 2747-2764. DOI: [10.1021/acsearthspacechem.9b00244](https://doi.org/10.1021/acsearthspacechem.9b00244)
7. Korol, R; Bou-Rabee, N; Miller*, T F Cayley modification for strongly stable path-integral and ring-polymer molecular dynamics. *J. Chem. Phys.* **2019**, 151 (12), 124103. DOI: [10.1063/1.5120282](https://doi.org/10.1063/1.5120282)
6. Korol, R; Segal*, D Machine Learning Prediction of DNA Charge Transport. *J. Phys. Chem. B* **2019**, 123 (13), 2801-2811. DOI: [10.1021/acs.jpcc.8b12557](https://doi.org/10.1021/acs.jpcc.8b12557)
5. Korol, R; Segal*, D From Exhaustive Simulations to Key Principles in DNA Nanoelectronics. *J. Phys. Chem. C* **2018**, 122 (8), 4206-4216. DOI: [10.1021/acs.jpcc.7b12744](https://doi.org/10.1021/acs.jpcc.7b12744)
4. Korol, R; Kilgour, M; Segal*, D ProbeZT: Simulation of transport coefficients of molecular electronic junctions under environmental effects using Büttiker's probes. *Comput. Phys. Commun.* **2018**, 224, 396-404. DOI: [10.1016/j.cpc.2017.10.005](https://doi.org/10.1016/j.cpc.2017.10.005)
3. Korol, R; Kilgour, M; Segal*, D Thermopower of molecular junctions: Tunneling to hopping crossover in DNA. *J. Chem. Phys.* **2016**, 145 (22), 224702. DOI: [10.1063/1.4971167](https://doi.org/10.1063/1.4971167)
2. Longobardi, L E; Zatsepin, P; Korol, R; Liu, L; Grimme, S; Stephan*, D W Reactions of Boron-Derived Radicals with Nucleophiles. *J. Am. Chem. Soc.* **2016**, 139 (1), 426-435. DOI: [10.1021/jacs.6b11190](https://doi.org/10.1021/jacs.6b11190)
1. Korol*, R V; Yanchuk, O M; Marchuk, O V; Orlov, V F; Moroz, I A; Vyshnevskiy, O A Size Stabilizers in Two-electrode Synthesis of ZnO Nanorods. *Phys. Chem. Solid St.* **2021**, 22 (2), 380-387. DOI: [10.15330/pcss.22.2.380-387](https://doi.org/10.15330/pcss.22.2.380-387)

PRESENTATIONS AND AWARDS	Quantum Thermodynamics and Decoherence conference , Binghamton University	2026
	<u>Invited talk</u> : “A window to Earth’s past with the help of theoretical chemistry”	
	Canadian Chemistry Conference and Exhibition , Toronto, Ontario	2026
	Contributed talk: “How NOT to build control-target gates in semiconductor quantum dots and beyond”	
	Physical, Theoretical and Computational Chemistry (PTC) General Session, Toronto, Ontario	2026
	Session chair	
	CQIQC Seminar, University of Toronto, Toronto, Ontario	2026
	<u>Invited talk</u> : “How NOT to build control-target gates in semiconductor quantum dots and beyond”	
	CERMM Seminar, Concordia University, Montreal, Quebec	2026
	<u>Invited talk</u> : “Quantum Effects in Complex Systems”	
	“Lunch and Learn” series, Department of Chemistry, Université de Sherbrooke, Quebec	2026
	Talk: “Quantum Effects in Complex Systems”	
	CCQS Seminar at the Physics department, University of Rochester, Rochester, New York	2025
	Talk: “How NOT to build control-target gates in semiconductor quantum dots and beyond”	
	Summer of Molecular Quantum Information Science Seminars Series	2025
	Talk: “Opening quantum dynamics on gate-defined quantum dots with RLC circuits”	
	Rochester Conference on Coherence and Quantum Science, Rochester, New York	2025
	Poster: “Hybrid singlet-triplet qubits enable fault-tolerant single-pulse CNOT gate”	
	American Conference on Theoretical Chemistry, Chapel Hill, North Carolina	2024
	Poster: “Analog Simulation of Open Quantum Dynamics”	
	Gray-Hill lecture at the Occidental college, Los Angeles, California.	2023
	<u>Award talk</u> : “A window to Earth’s past with the help of theoretical chemistry”	
	Canadian Chemistry Conference and Exhibition , Calgary, Canada	2022
	Contributed talk: “Accurate quantum statistics from improved path-integrals in imaginary time”	
	Molecular Science Mini-meeting , Montreal, Canada	2022
	Poster: “Dimension-free ring-polymer molecular dynamics”	
	ACS Spring meeting , San Diego, California	2022
	Poster: “Accurate quantum statistics from improved path-integrals in imaginary time”	
	Geological and Planetary Sciences seminar at Caltech, Pasadena, California	2022
	<u>Invited talk</u> : “ D and ^{13}C exchange equilibria using Path-Integral Monte-Carlo”	
	Berkeley Statistical Mechanics Meeting	2020
	Poster: “Cayley modification for strongly stable path-integral molecular dynamics”	
	CECAM BioMolecular Electronics Conference , Madrid, Spain	2018
	Poster: “Principles of Charge Transport in DNA: from extensive simulations to neural networks”	
	28 th Canadian Symposium on Theoretical and Computational Chemistry, Windsor, Canada	2018
	<u>Poster prize</u> : “Charge transport in DNA: From comprehensive simulations to key principles”	
	100 th Canadian Chemistry Conference , Toronto, Canada	2017
	<u>Poster prize</u> : “Tunneling to Hopping Crossover in Thermopower of DNA Molecular Junctions”	
	Chemical Biophysics Symposium , Toronto, Canada	2017
	Contributed talk: “DNA Molecular Junctions: Tunneling to Hopping Crossover”	

33rd [Symposium](#) on Chemical Physics, Waterloo, Canada

2017

Contributed talk: “Probing mechanisms of charge transport in DNA with Landauer-Büttiker formalism”

45th Southern Ontario Undergraduate Student Chemistry [Conference](#), Toronto, Canada

2017

1st prize talk: “Tunneling to Hopping Crossover in DNA & DNA-like molecular junctions”

SERVICE	Departmental Graduate Studies Committee (DGSC)	2026–present
	Student monitoring committees (3 students)	2026–present
	Student Representative at the Chemistry Department Advisory Committee	2016–2017
	2 nd year representative at the Chemistry student union	2016–2017
	Board member of the <i>Chemistry Connections</i> student group	2015–2016
MEMBERSHIP	Canadian Society for Chemistry (CSC)	
	Physical, Theoretical and Computational Chemistry (PTC) Division of the CSC	
	Canadian Association of Theoretical Chemists (CATC)	
	Centre for Research in Molecular Modeling (CERMM)	
	Centre for the Study of Advanced Materials for Batteries and Renewable Energy Storage Systems of the Université de Sherbrooke (CÉMAPROVUS)	
COMMUNITY VOLUNTEER INITIATIVES	Spearheaded and coordinated humanitarian supplies shipment to Ukraine 📦 from Caltech campus and beyond	2022–2023
	Volunteer at the Nova Ukraine non-profit, Stanford, California	2021–2023
	Website development, established and coordinated partnership with <i>Teach for Ukraine</i>	
	Volunteer at the Teach for Ukraine nonprofit, Kyiv, Ukraine	2021–2022
	Recruited and interviewed candidate teachers at the remote interview stage	
	International student orientation leader	2019, 2020
	“Big sibling” mentor for the incoming graduate students at Caltech	2019, 2020
	Science outreach program volunteer through Caltech Y	2018–2020
	High-school tutoring with CAUSE Tutoring non-profit	2018–2019
TEACHING	University of Toronto Peer Tutoring group tutor	2015–2018
	<i>Course development:</i> Computational chemistry labs, Chem3 at Caltech	2022
	Focus on structure-function relations and the dangers of approximations.	
	High School Teacher, Rotman Arts and Science School, Vaughan, Canada	2020–2022
	Academic stream, grade 11 and 12 Chemistry, Grade 10 Science.	
	Student placed 3rd in Vaughan, top 200 in Canada at the Avogadro chemistry contest	
	International Chemistry Olympiad Coach	
	Canadian National Team (4 students) – 2 bronze, 1 silver medals	Spring 2018
	Ukrainian National team (1 student) – bronze medal	Spring 2014
PEER REVIEW	Private tutoring of Chemistry, Physics and Math	2014–2018
	High school students: accepted to University college (UK), Columbia University (USA) and others	
	Chemical Biology summer school, Lutsk, Ukraine	Summer 2015
	Designed problems and experiments to help high school students master key concepts in chemistry	
	ACS journals: ACS Physical Chemistry Au	
	APS journals: Physical Review Letters, Physical Review A, B, E and Research	
	Elsevier journals: Organic Geochemistry, Chemical Geology	
	RSC journals: Physical chemistry chemical physics	
	Wiley journals: Rapid communications in mass spectrometry	

SUMMER SCHOOLS AND WORKSHOPS	Condensed Phase Dynamics Workshop at TSRC (Virtual)	2020
	Theoretical Chemistry School at TSRC , Telluride, Colorado	2019
	Weizmann Institute of Science, Rehovot, Israel Kupcinet-Getz Scholar at Rubtchinski lab	2015
PAST EMPLOYMENT	High School Teacher, Rotman Arts and Science School, Vaughan, Canada	2020–2022
	Research Assistant, Department of Linguistics, University of Toronto Heritage language variation and change project	2016–2018
EXTRA CURRICULARS	Rock climbing, weightlifting	since 2014
	Guitar, base	since 2012
	Ukrainian folk dance	since 2004
LANGUAGES	Fluent in Ukrainian, English & Russian	
COMPUTER LANGUAGES	Python, C++, MATLAB, FORTRAN, Mathematica, Bash; Web development (PHP & django, HTML, CSS, JS)	